TRIASHA SARKAR

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Machine Learning Engineer with experience in scientific ML, model evaluation, retrieval, time-series data, and ML systems. Background in aerospace engineering, simulation, and uncertainty.

PROFESSIONAL EXPERIENCE

Graduate Research Assistant under Prof. Dimitri Mavris | Georgia Tech ASDL | May 2025 – Aug 2026

- Built and tested Python models for projects in simulation, airline safety, and sustainable aviation; shared reproducible analyses with faculty and sponsors.
- Reviewed assumptions and limits with aerospace experts and documented each workflow so other researchers could rerun it.

Machine Learning Engineer | Rolls-Royce | Jul 2023 – Apr 2025

- Developed Python pipelines to flag unusual engine behavior, support maintenance planning, and combine sensor streams sampled at different rates.
- Mapped certification requirements to data and model checks, then reviewed results with lifecycle engineers before use.

Data Science Intern | Rolls-Royce DataLabs | May 2021 – Jul 2021

- Cleaned aircraft-engine sensor data, built features, and compared prediction and anomaly models; summarized recurring failure patterns for engineers.

TECHNICAL PROJECTS

EdgeGenBench | Real-flight ML and on-device inference | Aug 2026 – Present

- Trained a flight-anomaly model on NASA DASHlink recordings and tested it on 17,780 approaches from held-out aircraft. Its 0.738 macro F1 missed the release target.
- Exported the model to ONNX and tested corrupted sensor inputs; prediction agreement stayed above 99.55% across the tested cases.

IntegrityBench | Moderation evaluation and release controls | Aug 2026 – Present

- Built controlled evaluation and release checks for changing moderation policies, keeping separate candidates and frozen benchmark results distinct from deployment claims.

AeroSynth-Eval | Synthetic inspection-image evaluation | Aug 2026 – Present

- Evaluated synthetic aircraft-inspection image data with reproducible splits; reported where results were strong, limited, or still pending validation.

NewsLens | Recommendation and real-time search | Jul 2026 – Present

- Used chronological splits to keep future clicks out of training. The recommender scored 0.366 NDCG@10, but the uncertainty interval still included no improvement.
- Built a Go, Kafka, PostgreSQL, and FastAPI ingestion path for new articles. In a 500-event run, publish p99 was 44 ms and search freshness p95 was 79 ms.

Equity Backtest | Walk-forward signal evaluation | Jul 2026 – Sep 2026

- Built a cross-sectional equity-signal backtest with trading costs and safeguards against common look-ahead and selection biases.

Atlanta Mobility Resilience Digital Twin | Independent project | 2026 – Present

- Built an OSMnx and MARTA GTFS graph-analysis workflow covering 50 Census-tract origins, 101 essential-service destinations, and 216,659 estimated residents; the planned accessibility validation cycle is still pending.

Silent Failure Detection in Rocket-Motor Simulations | AE8900 Special Problems Coursework | May 2026 – Jul 2026

- Traced silent failures in a 15,120-case rocket-motor study to an uncapped convergence loop; a leakage-safe classifier reached 96.8% accuracy on unseen geometries.

Surrogate Model Learning | Reliability under design shift | Feb 2026 – Oct 2026

- Evaluated Gaussian-process and conventional surrogates on public airfoil, energy, and concrete data using grouped splits, multi-seed analysis, split-conformal intervals, and distance-to-training-domain guards.
- An airfoil Gaussian process reached R^2 0.8145 on a physically grouped split; a 10-seed mean of 0.8662 ± 0.0680 exposed split sensitivity. Nominal 90% intervals did not retain 90% coverage under design shift.

AIRFAANS | Extension of AE6394 Coursework Project | Jan 2026 – Apr 2026

- Compared an MLP, MeshGraphNet-style GNN, and point neural operator across three matched seeds and 200 AirfRANS interpolation meshes per model; MeshGraphNet had the lowest mean field and drag error.

- Used simulation-level splits and train-only normalization; matched OOD, uncertainty, and active-learning studies are in progress, with no results reported yet.

AeroRAG-X | Independent extension of Project HERO | Retrieval, evaluation, and ML systems | Oct 2025 – Present

- Extended Project HERO independently into a technical-knowledge system over 3,233 NASA report sections using hybrid retrieval, reranking, evidence checks, controlled citations, and containerized FastAPI services; retained negative experimental findings rather than claiming every change helped.

Project EAGLE | RIC Enterprise-sponsored Vehicle Grand Challenge | Georgia Tech ASDL | Sep 2025 – May 2026

- Developed sizing and synthesis tooling, passenger layout, AutoCAD cabin geometry, and requirements for the Eco-friendly Airborne Global Luxury Ekranoplan.

Project GREEN TEA | U.S. Endowment-sponsored Systems of Systems Grand Challenge | Georgia Tech ASDL | Sep 2025 – Present

- Developed a techno-economic framework for forest-residue sustainable aviation fuel and bioenergy, including jobs, CO2 impacts, a GREET surrogate, and forest-health modeling.

Project HERO | Delta Air Lines-sponsored source-aware retrieval for airline safety | Georgia Tech ASDL | May 2025 – Aug 2026

- Developed source-aware retrieval and evaluation methods for airline-safety analysis.

Supersonic Business Jets and Noise Mitigation | Sponsored by Rolls-Royce, Seneca, and DLR | Cranfield University | Sep 2022 – Feb 2023

Studied geometry, noise mitigation, high-speed aerodynamics, and design trade-offs.

Thermal System Design Engineer Group Project | Sponsored by GKN Aerospace | Cranfield University | Sep 2022 – Feb 2023

Worked in a multidisciplinary team on thermal-system architecture, requirements, sizing, and design trade-offs for an aerospace application.

Bird-Strike Impact Analysis | SRM Institute of Science and Technology | Jun 2020 – Jun 2021

Used ANSYS to compare aircraft nose-cone material choices for bird-strike impact.

Ornithopter Aerodynamics | SRM Institute of Science and Technology | Sep 2019 – May 2020

Used CFD to examine ornithopter wing area and load factor.

TECHNICAL SKILLS

ML and scientific ML: PyTorch, PyTorch Geometric, scikit-learn, graph neural networks, neural operators, surrogate modeling, uncertainty quantification, Transformers, RAG, BM25, dense retrieval, reranking

Systems: Python, Go, SQL, Kafka, FastAPI, Docker, Kubernetes, GitHub Actions, PostgreSQL/pgvector, Prometheus, OpenTelemetry, ONNX, Core ML, Qualcomm QNN

Scientific computing: NumPy, SciPy, pandas, CFD meshes, NetworkX, OSMnx, statistical inference, design of experiments

LEADERSHIP AND HACKATHONS

- Graduate Chair | Women of Aeronautics and Astronautics, Georgia Tech | Aug 2025 – Aug 2026
- Organising Team Member | Aarush Hackathon Event, SRM University | Jun 2019 – Jul 2019
- Student Volunteer | Aarush, SRM University | Aug 2018 – Jan 2019
- Team Participant | 36-Hour SRM Hackathon, SRM University | Sep 2019

  Worked in a five-member team on a sustainable urban-transport concept for densely populated Indian cities, comparing travel time, fuel use, cost, and CO2 emissions.

EDUCATION

M.S., Aerospace Engineering | Georgia Institute of Technology, Atlanta, GA | Aug 2026

GPA: 3.7/4.0

M.Sc., Aerospace Vehicle Design | Cranfield University, United Kingdom | Feb 2023

GPA: 3.6/4.0 | First Class Honours | Thesis: geometry optimization under FAA constraints

B.Tech., Aerospace Engineering | SRM Institute of Science and Technology, India | Jun 2021

GPA: 3.8/4.0 | First Class Honours